

Course Registration Database — Analytical Report

CourseRegistrationDB • Spring 2026 Term • Northeastern University

This report presents an analytical overview of the CourseRegistrationDB database for the Spring 2026 term at Northeastern University. The database tracks 10 course sections across five departments, managing enrollment, waitlists, prerequisites, scheduling, and student records. The analysis that follows is organized into seven sections, each examining a distinct operational dimension of the registration system: enrollment capacity, departmental distribution, waitlist dynamics, prerequisite dependencies, scheduling patterns, room utilization, and student demographics. The objective is to identify areas of strong demand, operational bottlenecks, and opportunities for resource optimization.

10

Courses Offered

10

Active Sections

435

Total Enrolled

530

Total Capacity

82.1%

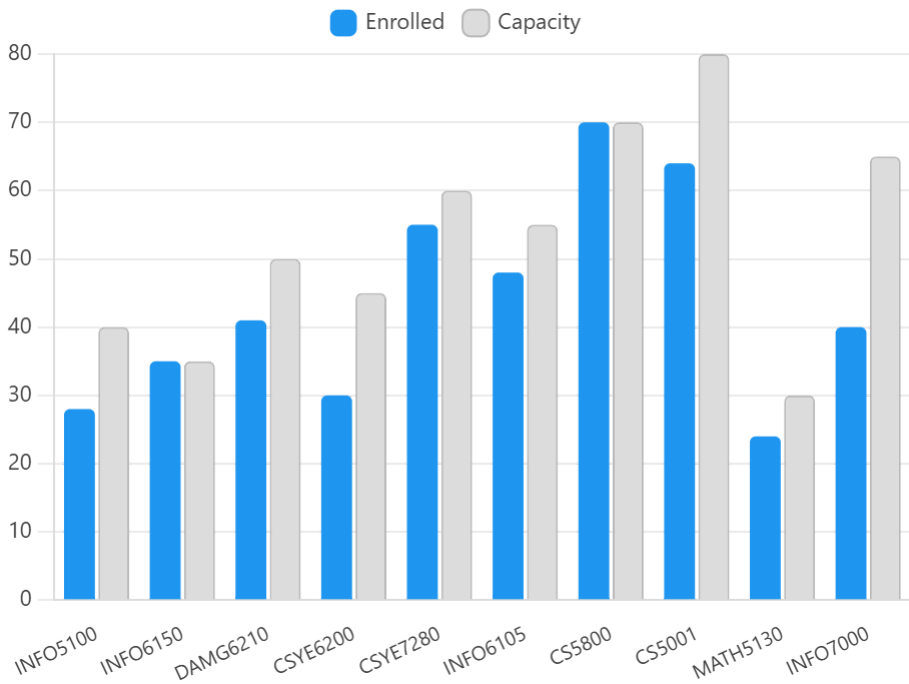
Avg Fill Rate

At a glance, the Spring 2026 term features 10 course sections with a combined capacity of 530 seats, of which 435 are currently occupied — yielding an average fill rate of 82.1%. This figure indicates healthy overall demand, though individual sections vary considerably, from 61.5% (INFO7000) to 100% (INFO6150, CS5800). Two sections have been marked Closed, and both carry active waitlists. These high-level metrics set the stage for the more detailed analyses below.

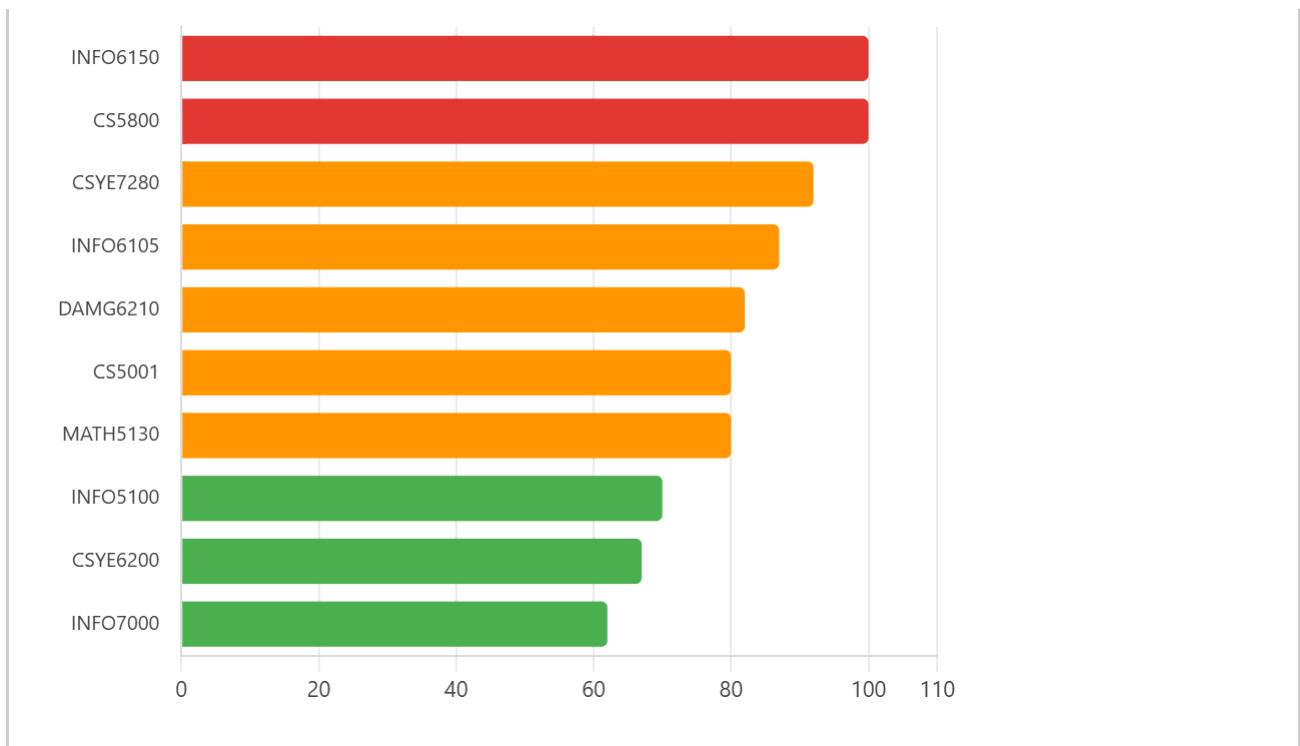
Section 1. Enrollment & Capacity Analysis

Understanding the relationship between enrollment and capacity is essential for identifying oversubscribed courses and underutilized sections. The two charts below compare absolute enrollment counts against section capacity (left) and rank all sections by their fill rate percentage (right). A fill rate above 90% typically signals that a section is at risk of turning students away, while rates below 70% may indicate weak demand or scheduling conflicts.

Enrollment vs. Capacity by Section



Fill Rate by Section (Ranked)



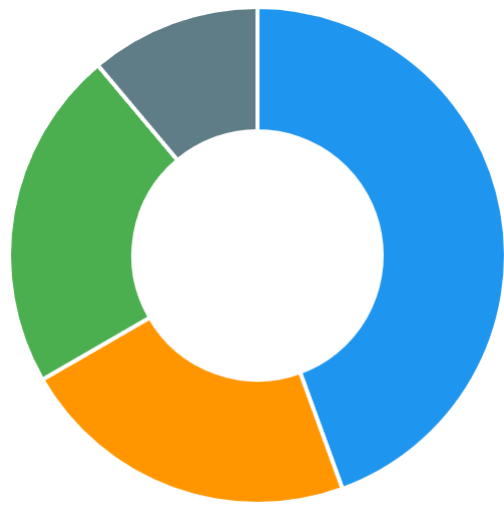
The data reveals a clear bifurcation. At the top end, INFO6150 (Web Design) and CS5800 (Algorithms) have reached 100% capacity and are officially closed to new enrollment. CSYE7280 (UX Design and Testing) follows closely at 91.7%, and INFO6105 (Data Science Methods) at 87.3% — both approaching functional saturation. These four sections collectively account for 223 of the 435 enrolled students (51.3%), suggesting that demand is concentrated in a handful of high-interest courses.

At the lower end, INFO7000 (Capstone) stands at only 61.5% capacity and CSYE6200 (Object-Oriented Design) at 66.7%. The Capstone course's relatively low enrollment is likely structural rather than a demand issue: it requires three prior courses as prerequisites (INFO5100, INFO6150, and DAMG6210), which naturally limits the eligible student pool. CSYE6200's lower fill rate may reflect its stricter prerequisite grade requirement of B- in CS5001, compared to the C required by most other courses.

Section 2. Department & Instructor Distribution

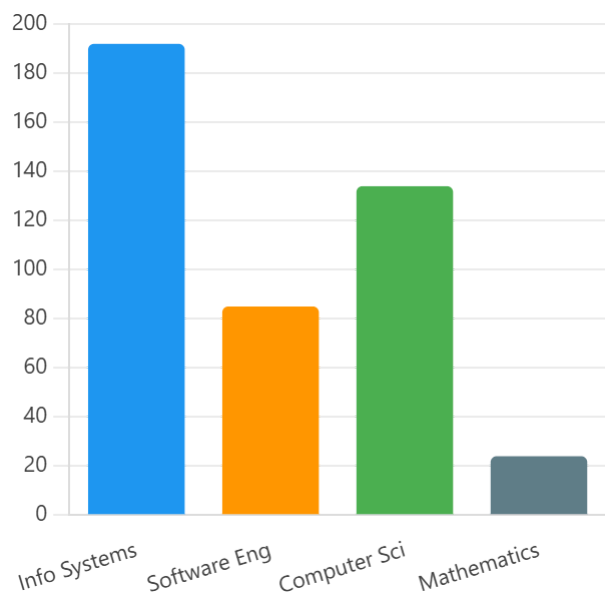
The Spring 2026 course offerings span four departments. Examining the distribution of sections and enrollment across these departments reveals where institutional resources are concentrated and where faculty workloads may be uneven.

Sections by Department

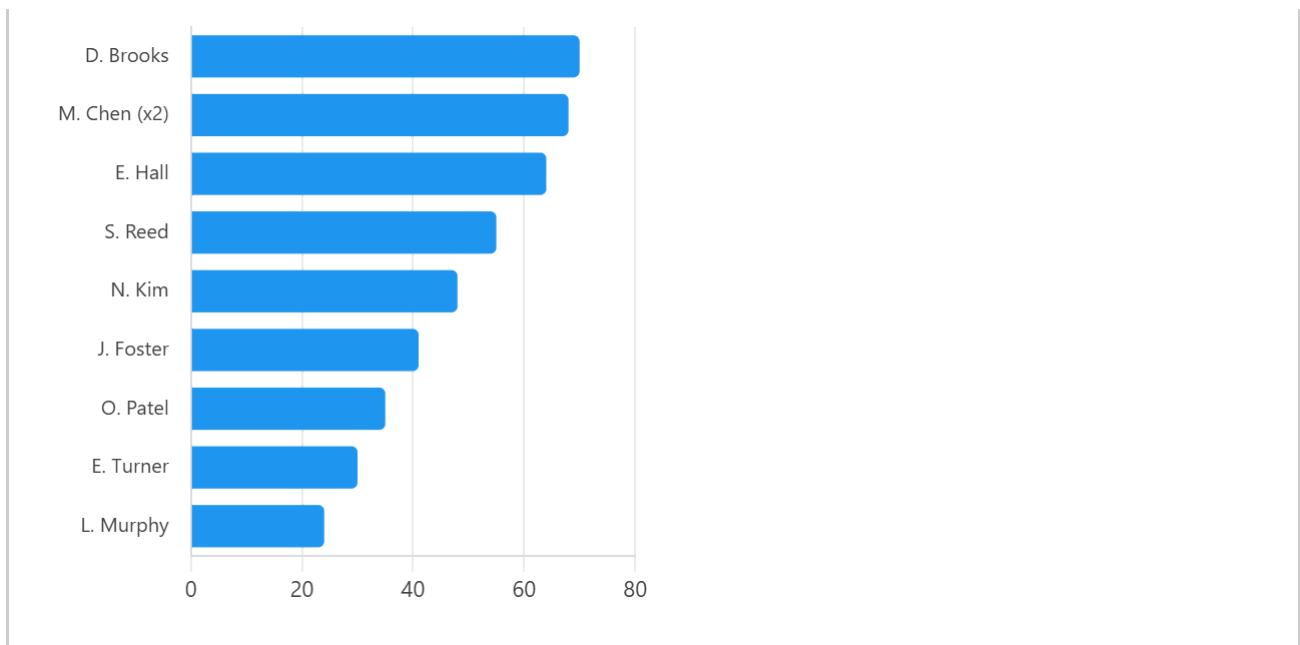


Information Systems (4) Software Engineering (2)
Computer Science (2) Mathematics (1)

Enrollment by Department



Instructor Teaching Load



Information Systems dominates the term with 4 sections and 192 enrolled students, reflecting its central role in the graduate curriculum. Computer Science offers only 2 sections but enrolls 134 students — an average of 67 per section — the highest per-section density of any department. This suggests that Computer Science courses face the greatest demand pressure relative to their supply. Software Engineering (2 sections, 85 students) and Mathematics (1 section, 24 students) round out the offerings.

From an instructor workload perspective, Daniel Brooks carries the largest single-section load with 70 students in CS5800 (Algorithms). Maya Chen is notable for teaching two sections — INFO5100 and INFO7000 — totaling 68 students across both. Ethan Hall manages 64 students in CS5001, the foundational gateway course. At the other extreme, Liam Murphy teaches only 24 students in MATH5130, though this is constrained by Hayden 150's 30-seat capacity rather than low demand (the section is 80% full). Faculty load balancing may warrant attention in future terms, particularly for instructors managing large sections without teaching assistant support.

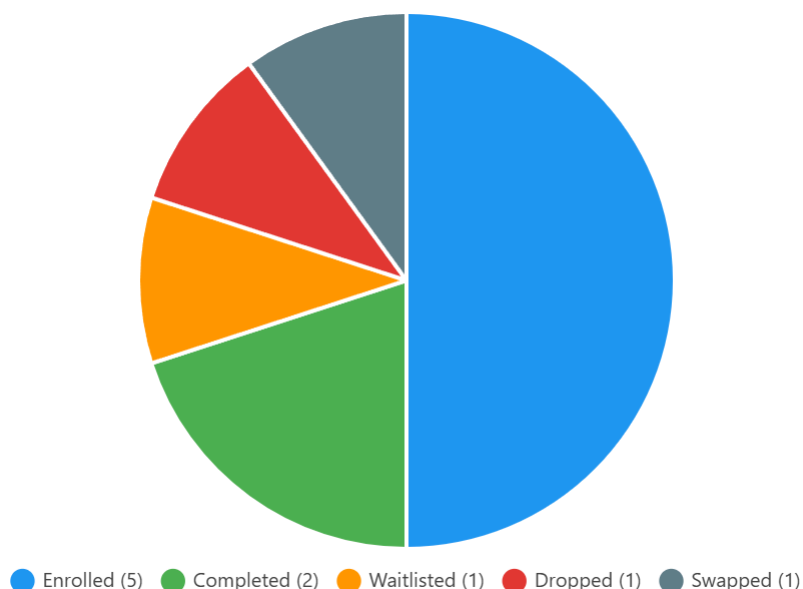
Section 3. **Waitlist & Registration Status**

When a section reaches full capacity, students are placed on a waitlist. Analyzing waitlist depth provides a direct measure of unmet demand, while the overall distribution of registration statuses reveals the operational health of the enrollment process.

Waitlist Pressure on Closed Sections



Registration Status Breakdown



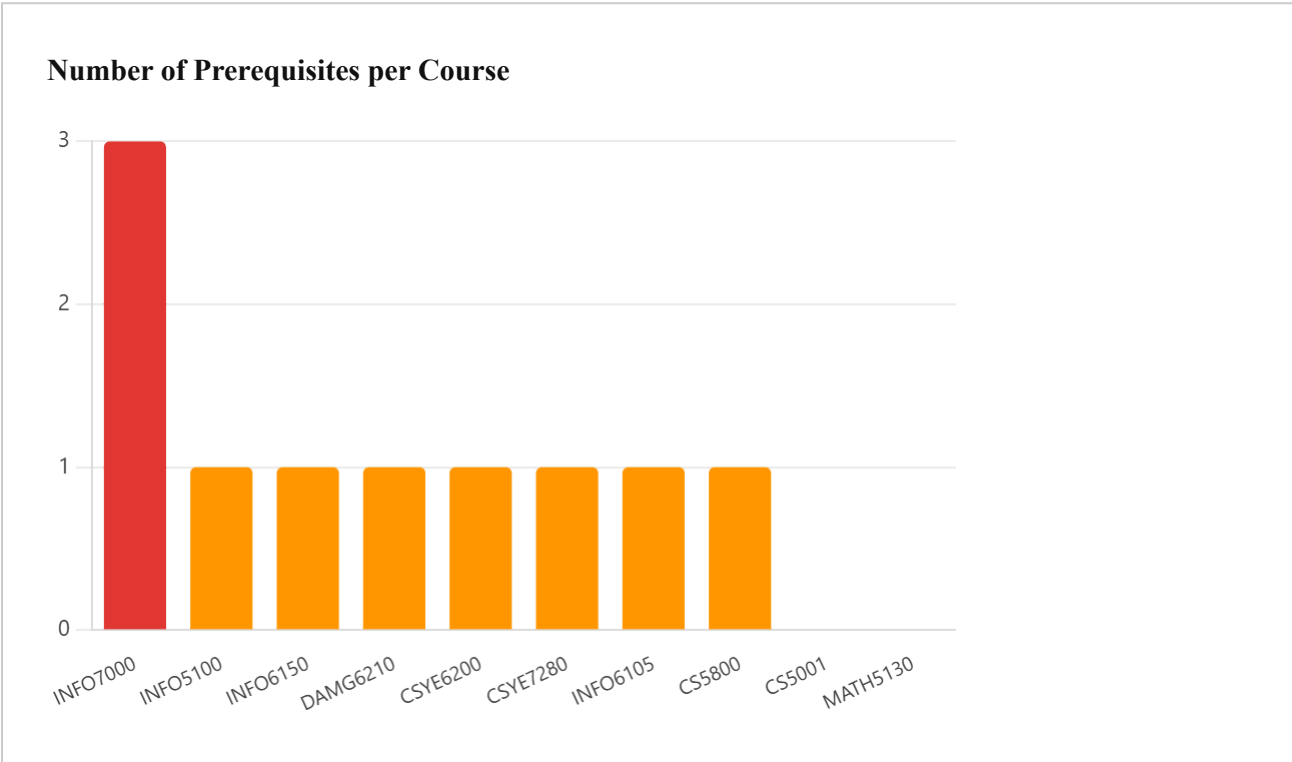
Both closed sections — INFO6150 and CS5800 — carry identical waitlist depths of 5 students each. For INFO6150, with a capacity of only 35, this represents a 14.3% overflow relative to available seats. For CS5800, the 5 waitlisted students represent a 7.1% overflow against its 70-seat capacity. In both cases, the waitlist includes students at various statuses: Active (awaiting a seat), Offered (a seat has been made available), and Removed (voluntarily withdrew). The presence of sustained Active waitlists suggests that simply waiting for attrition will not resolve the demand gap — additional sections or larger classrooms should be considered.

Across all 10 registration records in the sample, 5 are actively enrolled (50%), 2 have completed their courses (20%), 1 is waitlisted (10%), 1 has dropped (10%), and 1 has been swapped to another section (10%). The 10% drop rate, while based on a small sample, is worth monitoring over time. The dropped

registration belongs to Kevin Lee (student_id 5), who is also the only student on academic probation — suggesting a potential correlation between academic standing and enrollment stability.

Section 4. Prerequisite Dependency Analysis

Prerequisites shape the flow of students through the curriculum. A course with many prerequisites will naturally have a smaller eligible enrollment pool, while a course that serves as a prerequisite for many others becomes a critical pipeline bottleneck if its capacity is constrained.



Prerequisite Dependency Map		
Course	Prerequisite	Min. Grade
INFO5100	CS5001	C
INFO6150	CS5001	C
DAMG6210	CS5001	C
CSYE6200	CS5001	B-
CS5800	CS5001	B
CSYE7280	INFO6150	C

Course	Prerequisite	Min. Grade
INFO6105	MATH5130	C
INFO7000	INFO5100, INFO6150, DAMG6210	C (each)

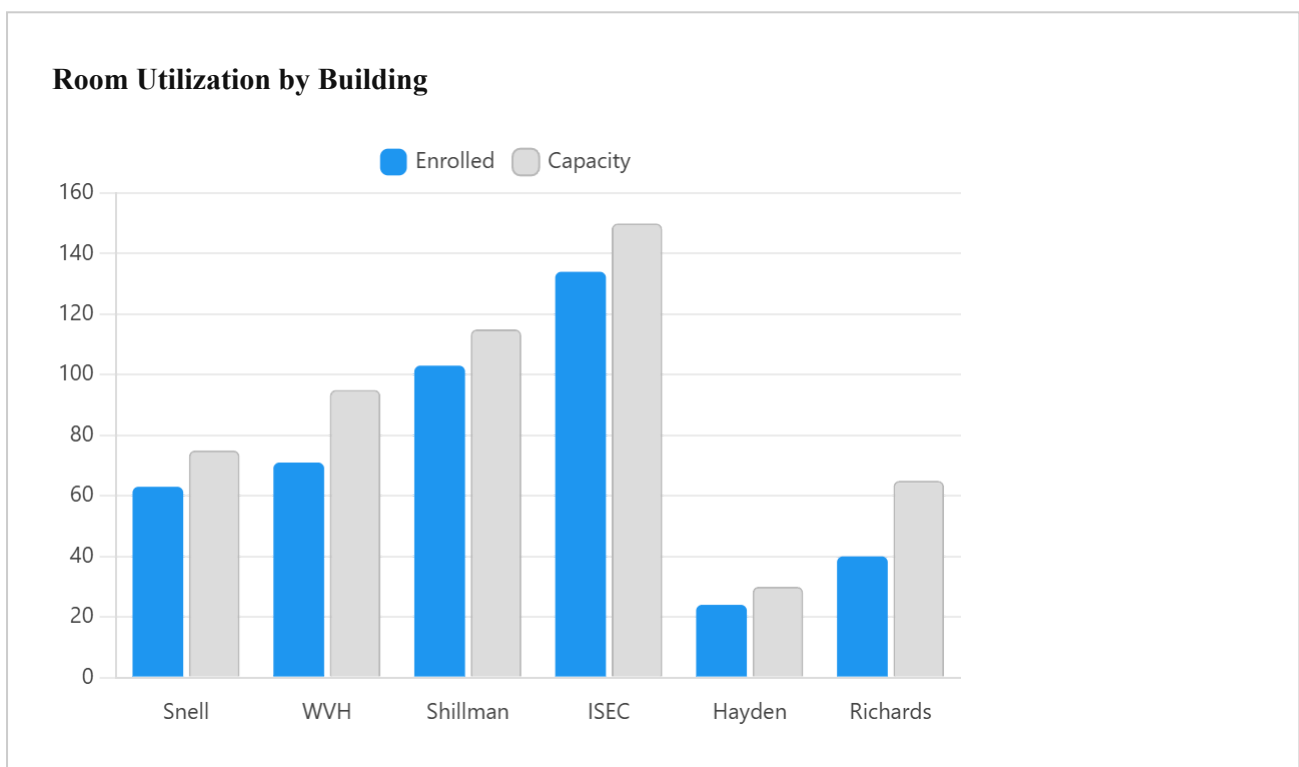
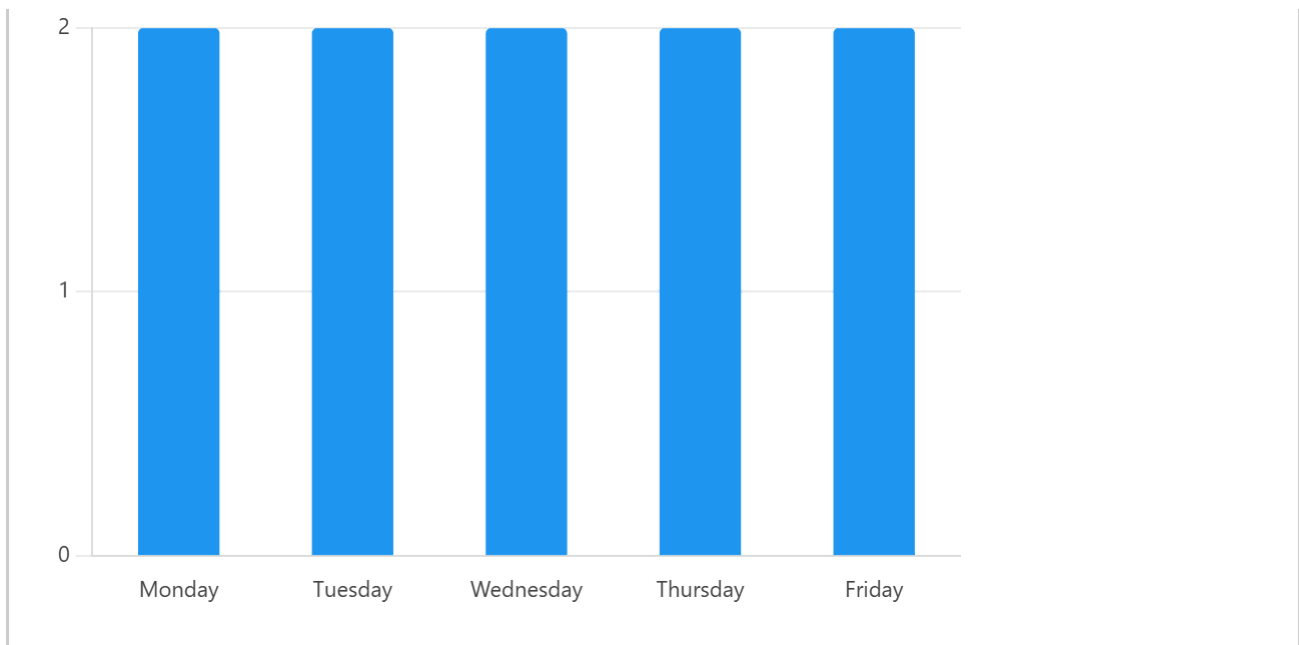
The prerequisite structure reveals CS5001 (Intensive Foundations of Computer Science) as the single most critical course in the curriculum. Five other courses — INFO5100, INFO6150, DAMG6210, CSYE6200, and CS5800 — all require it as a prerequisite. This means that any capacity constraint or scheduling issue in CS5001 would have cascading effects across the entire program. Currently, CS5001 is at 80% capacity (64 of 80 seats filled), leaving 16 available seats. While this buffer appears adequate for the current term, the course's centrality warrants proactive capacity planning for future terms.

MATH5130 (Statistics for Data Science) serves as the sole prerequisite for INFO6105 and is the only non-CS5001 gateway course. Its small classroom (Hayden 150, 30 seats) could become a bottleneck as demand for data science coursework grows. INFO7000 (Capstone) sits at the terminus of the prerequisite chain, requiring completion of three courses — which explains its comparatively low fill rate of 61.5%. Students must pass INFO5100, INFO6150, and DAMG6210, all with a minimum grade of C, before becoming eligible. This three-course prerequisite chain, combined with the fact that INFO6150 is already closed, could further constrain Capstone enrollment in upcoming terms.

Section 5. Schedule & Room Utilization

Efficient use of classroom space and balanced scheduling across the week are important operational objectives. This section examines how sections are distributed across weekdays and how effectively building space is being utilized.

Section Distribution by Day of Week



The schedule is evenly distributed at exactly 2 sections per day, Monday through Friday. All sections follow a uniform 100-minute meeting block, consistent with a graduate seminar format. No sections are offered on Saturday, leaving that day available for makeup sessions or special events. The even distribution eliminates concerns about peak-day congestion, though it also means there is no flexibility to absorb additional sections on any given day without extending into new time slots.

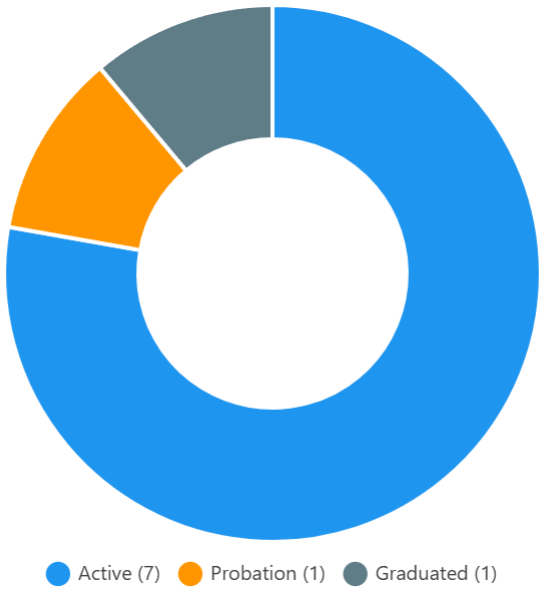
Room utilization varies by building. ISEC accounts for the largest share of total capacity (150 seats across two rooms) and hosts 134 enrolled students — an overall utilization rate of 89.3%. Shillman Hall follows with 115 seats and 103 students (89.6%), making it the most efficiently utilized building. Snell Library's two rooms offer 75 seats total with 63 students (84.0%). At the other end, Richards Hall is the least utilized at 61.5% (40 students in a 65-seat room), though this is attributable to the Capstone course's prerequisite-

limited enrollment rather than a room allocation issue. Hayden Hall operates at 80.0% but its small 30-seat capacity makes it unsuitable for high-demand courses — a limitation that could matter as MATH5130 enrollment grows.

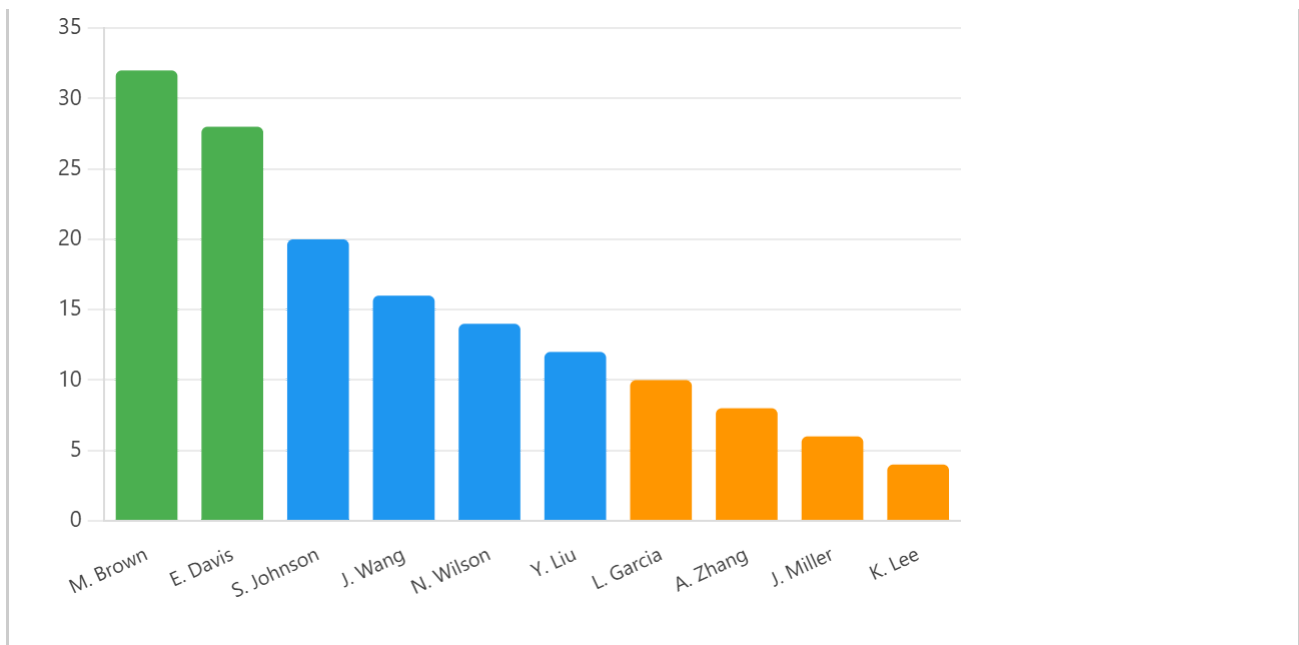
Section 6. Student Profile Analysis

The database contains records for 10 students at various stages of academic progress. Examining their status distribution and credit accumulation provides insight into the student population's overall health and identifies potential at-risk individuals.

Student Status Distribution



Credit Hours Completed per Student



Of the 10 students in the system, 7 hold Active status (70%), 1 is on Probation (10%), 1 has Graduated (10%), and 1 is classified as Inactive (0% — not present in this sample but defined in the schema). The high Active rate is expected for a current-term snapshot, but the presence of a Graduated student (Michael Brown, 32 credits) still carrying a registration record (Completed, grade B+ in CS5800) suggests that historical records are retained in the system, which is appropriate for transcript and audit purposes.

Credit hour accumulation ranges from 4 (Kevin Lee) to 32 (Michael Brown). The median is 13 credits, and the mean is 15.0 credits. Kevin Lee stands out as the only student on Probation and has the fewest completed credits at just 4 hours. His registration record shows a Dropped status with a grade of W (withdrawal), and he was also removed from the CSYE7280 waitlist. This pattern — low credits, probation status, course withdrawal — identifies him as the most at-risk student in the cohort and a candidate for academic advising intervention.

At the upper end, Sarah Johnson (20 credits, Active) is notable for having an elevated maximum credit hour allowance of 21, compared to the standard 18. This suggests she may have received an academic override, possibly reflecting strong academic performance. Emily Davis (28 credits, Active) is approaching graduation-level credit accumulation and may be in her final term of coursework.

Section 7. Complete Section Overview

The table below provides a comprehensive reference for all 10 course sections offered in Spring 2026, consolidating course codes, instructor assignments, room locations, meeting schedules, enrollment figures, and section status into a single view.

Course	Instructor	Room	Day	Time	Enrolled	Capacity	Fill %	Status
INFO5100	Maya Chen	Snell 101	Mon	9:00–10:40	28	40	70.0%	Open
INFO6150	Olivia Patel	Snell 202	Tue	11:00–12:40	35	35	100.0%	Closed
DAMG6210	James Foster	WVH 120	Wed	1:00–2:40	41	50	82.0%	Open
CSYE6200	Emma Turner	WVH 210	Thu	9:00–10:40	30	45	66.7%	Open
CSYE7280	Sophia Reed	Shillman 305	Fri	10:00–11:40	55	60	91.7%	Open
INFO6105	Noah Kim	Shillman 110	Mon	2:00–3:40	48	55	87.3%	Open
CS5800	Daniel Brooks	ISEC 420	Tue	3:00–4:40	70	70	100.0%	Closed
CS5001	Ethan Hall	ISEC 515	Wed	9:00–10:40	64	80	80.0%	Open
MATH5130	Liam Murphy	Hayden 150	Thu	1:00–2:40	24	30	80.0%	Open
INFO7000	Maya Chen	Richards 220	Fri	3:00–4:40	40	65	61.5%	Open

Section 8. Summary & Recommendations

This analysis of the CourseRegistrationDB for Spring 2026 yields several actionable findings. First, two sections (INFO6150 and CS5800) are at full capacity with active waitlists of 5 students each; the department should consider opening additional sections or relocating these courses to larger classrooms in future terms. Second, CS5001 serves as the prerequisite gateway for five downstream courses, making it a single point of failure in the curriculum pipeline — its capacity should be monitored closely and expanded proactively if enrollment trends upward. Third, the prerequisite chain leading to INFO7000 (Capstone)

creates a natural enrollment ceiling that currently limits its fill rate to 61.5%; this is structural and not necessarily a concern, but the closure of INFO6150 this term could reduce the Capstone-eligible pool in subsequent semesters.

On the student side, Kevin Lee's profile — low credit accumulation, probation status, and course withdrawal — flags him as an at-risk student who would benefit from early intervention. Faculty workload distribution is generally reasonable, though Daniel Brooks and Maya Chen carry notably heavier loads than their peers. Finally, classroom utilization is efficient overall (82.1% average), but the small capacity of Hayden 150 (30 seats) may become a constraint as demand for MATH5130 grows alongside the expanding data science curriculum.